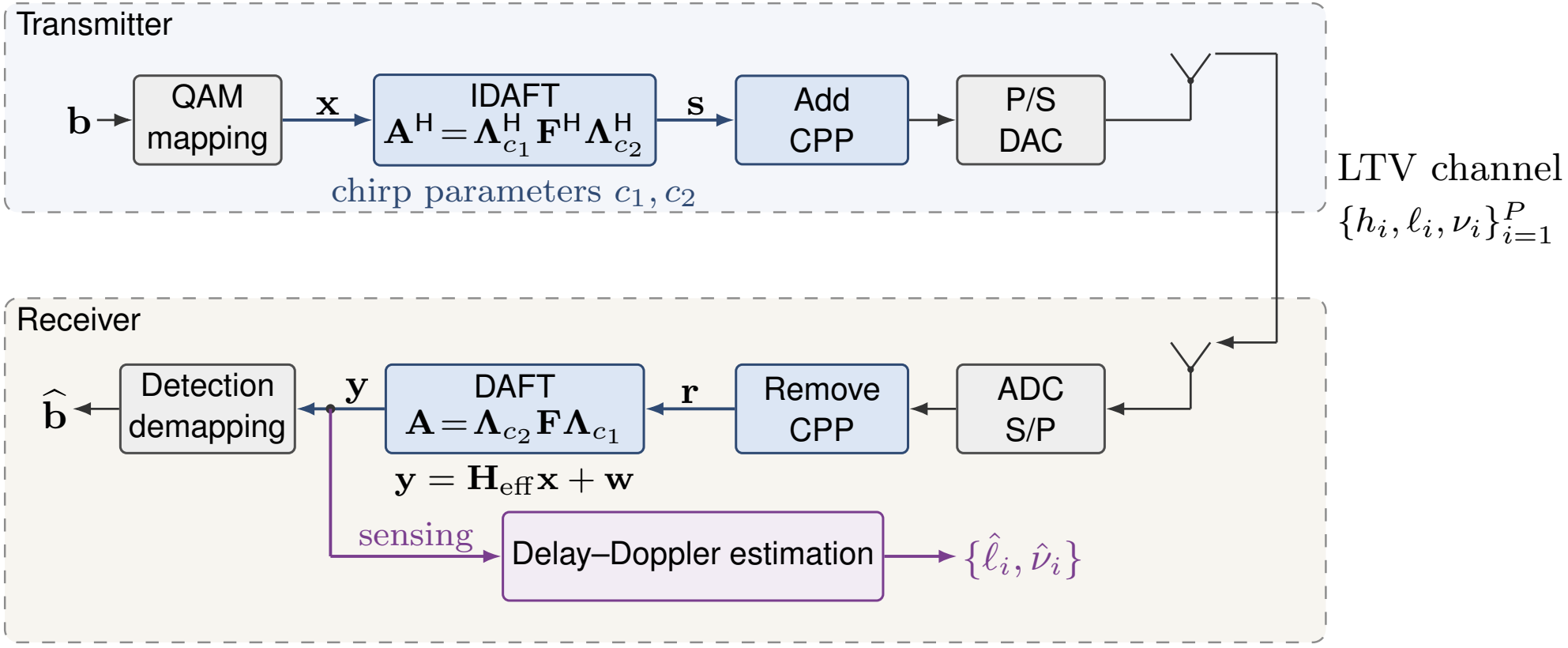

Transmitter
b
QAM mapping
x
IDAFT
$\mathbf{A}^{\mathsf{H}} = \mathbf{\Lambda}_{c_1}^{\mathsf{H}} \mathbf{F}^{\mathsf{H}} \mathbf{\Lambda}_{c_2}^{\mathsf{H}}$
s
Add CPP
P/S DAC
chirp parameters c_1, c_2
LTV channel
$\{h_i, \ell_i, \nu_i\}_{i=1}^{P}$
Receiver
$\widehat{\mathbf{b}}$
Detection demapping
y
DAFT
$\mathbf{A} = \mathbf{\Lambda}_{c_2} \mathbf{F} \mathbf{\Lambda}_{c_1}$
r
Remove CPP
ADC S/P
$\mathbf{y} = \mathbf{H}_{\text{eff}} \mathbf{x} + \mathbf{w}$
sensing
Delay–Doppler estimation
$\{\hat{\ell}_i, \hat{\nu}_i\}$